

E-COMMERCE BUSINESS ANALYTICS

Final Business Investigation & Recommendations Report

Analysis scope	Sales, profitability, products, customers, regions, discounts, returns and marketing
Data sources	Customers, Products, Orders, Order Items, Payments, Returns and Marketing
Final investigation	Part 9 – Business Investigation
Final recommendations	Part 10 – Final Recommendations

Prepared from the completed Python and Power BI analysis.

1. Executive Summary

The analysis supports management's statement that revenue growth is not translating into profitability and customer retention at the same rate. The major pressure points are loss-making products, product costs, discounting, customer retention risk, weaker regional revenue performance, product returns, and inefficient Facebook marketing.

The strongest opportunities are to improve product-level profitability, retain valuable customers, reduce avoidable returns, improve Central and East regional performance, control discounts, and shift marketing investment toward more efficient campaigns.

2. Dataset Overview

Dataset	Rows	Columns	Data quality summary
Customers	500	6	0 duplicates; 0 missing
Products	80	6	0 duplicates; 0 missing
Orders	2,199	4	0 duplicates; 0 missing
Order Items	5,534	13	Missing values detected; 1 duplicate in the raw check
Payments	2,200	6	0 duplicates; 0 missing
Returns	259	6	0 duplicates; 0 missing
Marketing	12	14	0 duplicates; 0 missing

The raw data-quality check identified missing values and a duplicate in Order Items. The project workflow uses the cleaned datasets for downstream analysis.

3. Statistical & Correlation Analysis

Mean revenue was \$19,909.58 and median revenue was \$8,946.63. Revenue had a standard deviation of \$740,100.86, indicating substantial variation in the revenue observations. Mean profit was \$3,298.70 and median profit was \$1,819.59.

Relationship	Correlation	Interpretation
Discount vs Profit	-0.242	Negative association; higher discounts are associated with lower profit
Cost Price vs Profit	-0.445	Stronger negative association; product cost pressure is important
Revenue vs Profit	0.587	Higher revenue is generally associated with higher profit
Profit vs Profit Margin	0.783	Strong positive association

4. Product Analysis

Top profitable products

- Product 010 — \$930,094.37 profit
- Product 061 — \$854,527.81 profit
- Product 068 — \$794,949.56 profit
- Product 014 — \$702,499.80 profit
- Product 001 — \$698,811.86 profit

Major loss-making products

- Product 004 — -\$41,028.01
- Product 042 — -\$38,954.02
- Product 045 — -\$33,242.70
- Product 026 — -\$29,890.85
- Product 059 — -\$29,470.96

The product analysis indicates that revenue volume alone does not guarantee profitability. Several products generate substantial sales while producing negative profit, making pricing, supplier cost and discount review a priority.

5. Customer Analysis & Segmentation

Segment	Customers	Total spending	Avg. recency
High-Value	177	\$20.11M	76 days
VIP	103	\$19.90M	64 days
At-Risk	88	\$8.45M	246 days
Regular	80	\$4.69M	85 days
Inactive	30	\$1.47M	451 days
Low-Value	16	\$0.42M	86 days

The retention concern is concentrated in 88 At-Risk and 30 Inactive customers. At-Risk customers have average spending of \$96,017.25, 3.78 average orders and 246 days of recency, making them a high-priority win-back opportunity.

6. Regional Analysis

Region	Revenue	Profit where confirmed
North	\$13.1M	—
South	\$12.5M	—
West	\$12.2M	—
East	\$8.8M	\$2.93M
Central	\$8.5M	\$2.91M

Central has the lowest regional revenue at \$8.5M, followed by East at \$8.8M. Based on the confirmed figures, Central and East warrant targeted investigation and improvement. The available figures do not establish a complete profit ranking for all five regions.

7. Discount & Returns Analysis

Discount has a negative correlation with profit of -0.242. Product 004 is a clear example of the risk: it has an average discount of approximately 15.24% while generating a loss of \$41,028.01. Discount limits should therefore be reviewed for low-margin products.

There are 259 recorded returns. Product 018 has the highest return count in the analysis with 8 returns. Other high-return products include Products 017, 009, 053, 029 and 059. High-return products should be investigated for quality, delivery, description and expectation issues.

8. Marketing Analysis

Channel	ROI	Conversion rate	CAC	Revenue
Instagram	2,173.98%	0.98%	\$60.15	\$3.07M
YouTube	1,717.45%	84.88%	\$12.16	\$3.32M
Google Ads	1,455.59%	9.76%	\$85.49	\$4.61M
Email	704.46%	5.01%	\$160.35	\$2.65M
Facebook	280.03%	2.76%	\$294.78	\$0.88M

Facebook is the least efficient channel in the supplied analysis, with the lowest ROI (280.03%), highest CAC (\$294.78) and lowest conversion rate (2.76%). Google Ads generated the highest revenue at \$4.61M. YouTube had the lowest CAC at \$12.16 and the highest reported conversion rate at 84.88%.

9. Part 9 – Business Investigation

1. What is causing the profitability problem?

Profitability is mainly affected by high product costs, excessive discounting and loss-making products. Discount vs profit correlation is -0.242, while cost price vs profit correlation is -0.445.

2. Which products are responsible?

The largest identified losses are Product 004 (-\$41,028), Product 042 (-\$38,954), Product 045 (-\$33,243), Product 026 (-\$29,891) and Product 059 (-\$29,471).

3. Which customer segments are responsible?

The main retention-risk segments are At-Risk Customers (88) and Inactive Customers (30), giving 118 customers requiring retention or reactivation attention.

4. Which regions are underperforming?

Central has the lowest revenue at \$8.5M, followed by East at \$8.8M. Confirmed profits are \$2.91M for Central and \$2.93M for East, so both regions require investigation.

5. Are discounts affecting profit?

Yes. The negative correlation of -0.242 indicates an association between higher discounting and lower profitability. High-discount, loss-making products should be reviewed.

6. Are returns affecting profitability?

There are 259 returns. Refunds can reduce realized profitability, so products with high return volumes should be investigated.

7. Which marketing channels are inefficient?

Facebook is the least efficient channel in the supplied analysis: 280.03% ROI, \$294.78 CAC and 2.76% conversion rate.

8. Which customers are at risk?

There are 88 At-Risk Customers. Their average spending is \$96,017.25, average orders are 3.78 and average recency is 246 days.

9. What are the biggest business opportunities?

Retain At-Risk customers, protect VIP and High-Value customers, fix loss-making products, reduce excessive discounting, improve Central and East, optimize Facebook spending and reduce product returns.

Overall conclusion

The investigation supports management's statement. Revenue growth is not translating into profitability and retention at the same rate. Profitability is being pressured by loss-making products, product costs and discounting, while retention is affected by At-Risk and Inactive customers.

10. Part 10 – Final Recommendations

1. Improve Loss-Making Products

Problem: Several products generate negative profit.

Supporting data: Product 004 lost \$41,028.01 and Product 042 lost \$38,954.02.

Recommended action: Review pricing, supplier costs and discount levels; increase prices or reduce costs where commercially feasible.

Expected business impact: Reduce product-level losses and improve overall profit margin.

2. Reduce Excessive Discounting

Problem: Higher discounts are associated with lower profit.

Supporting data: Discount-profit correlation is -0.242; Product 004 averages about 15.24% discount while making a loss.

Recommended action: Set discount limits for low-margin products and avoid deep discounts on loss-making items.

Expected business impact: Protect profit margins while maintaining sales volume.

3. Retain At-Risk Customers

Problem: A significant customer group is at risk of becoming inactive.

Supporting data: 88 At-Risk Customers have average recency of about 246 days.

Recommended action: Launch targeted win-back campaigns using personalized offers and reminders.

Expected business impact: Increase repeat purchases and improve retention.

4. Re-engage Inactive Customers

Problem: Inactive customers have very long periods since their last purchase.

Supporting data: 30 Inactive Customers have average recency of about 451 days.

Recommended action: Create a separate reactivation campaign with personalized promotions and product recommendations.

Expected business impact: Recover previously inactive customers and generate incremental revenue.

5. Improve Central Region Performance

Problem: Central has the lowest regional revenue.

Supporting data: Central generated \$8.5M revenue and \$2.91M confirmed profit.

Recommended action: Analyze local product demand and customer behavior and introduce targeted regional promotions.

Expected business impact: Increase regional sales and profit contribution.

6. Optimize Facebook Marketing Spending

Problem: Facebook is the least efficient channel in the supplied analysis.

Supporting data: ROI 280.03%, CAC \$294.78 and conversion rate 2.76%.

Recommended action: Reduce inefficient Facebook campaigns and reallocate part of the budget to better-performing campaigns.

Expected business impact: Lower acquisition cost and improve marketing ROI.

7. Reduce Product Return Problems

Problem: Returns create refund costs and can reduce realized profitability.

Supporting data: 259 total returns; Product 018 has the highest return count at 8.

Recommended action: Investigate high-return products for quality, delivery, product-description and expectation issues.

Expected business impact: Reduce refunds, improve customer satisfaction and protect profitability.

8. Protect High-Value and VIP Customers

Problem: High-value customers contribute significantly to business revenue and should be retained.

Supporting data: 177 High-Value Customers and 103 VIP Customers.

Recommended action: Introduce loyalty benefits, personalized offers and priority engagement.

Expected business impact: Increase customer lifetime value and reduce loss of high-spending customers.

11. Final Recommendation Summary

The business should focus on improving product profitability, controlling discounts, retaining valuable customers, reducing returns, improving weak regional performance and reallocating marketing spending toward efficient channels. These actions can improve profitability and customer retention while supporting sustainable revenue growth.

12. Final Business Conclusion

The overall analysis confirms that increasing revenue alone is not sufficient to deliver stronger business performance. The key management priority is to improve the quality of revenue by protecting margins, retaining high-value customers, reducing avoidable refunds and concentrating marketing investment on efficient channels.

The recommended actions provide a practical path from analysis to business improvement: fix loss-making products, control discounting, recover at-risk customers, strengthen Central and East, reduce return-related losses, and optimize the marketing budget.